Le réseau de transport d'électricité

MINES PARIS | PSL

Assessment of the potential of Earth observation data and deep convolutional neural networks to improve the estimation and forecast of the solar power production in France

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Context and overview

Not all PV are measured equally

Photovoltaic energy

Plants ($\geq 1\ MWp$)

Mesurable : RTE can access real time measurements of the production

Decentralized$< 1\ MWp$)

Small plants

Distributed decentralized

Observable : RTE can estimate the PV production

- Increasing installed capacity
- Ever greater impact on the grid
- ***Not observable***

Increasing uncertainty for load and supply balancing

Bridging the gap leveraging a regional PV model and a registry

We can fit a **regional PV Model** to get an accurate estimation of the decentralized PV
However, **we need statistics on the characteristics of PV systems**

Regional PV model

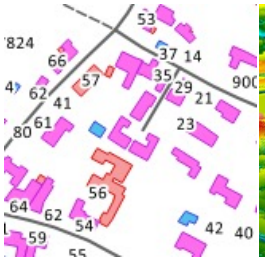
Distributed PV chatacteristics cannot be extrapolated from plants

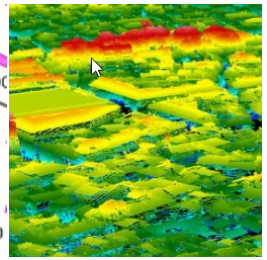
Construct a **registry** containing the following characteristics :

- **Localization**
- **Tilt**
- **Azimuth**
- **Surface area**
- **Installed capacity**

Leveraging deep learning and EO data to build the registry

Previous studies (Yu et al, 2018, Mayer et al, 2022) proved that **deep learning** and **Earth observation data** can be leveraged to **construct large scale PV registries**





Id	Loc.	Tilt	Azim	Inst. Cap.	Area
0	45°, 3°	30°	-10°	3 kWp	12 sqm
1	44°, 3°	45°	-30°	2,9 kWp	10 sqm
2	46°, 3°	10°	0°	30 kWp	100 sqm
			…		

1. Detection and segmentation of PV installations

➔ **Image classification and image segmentation**

2. Filtering and characteristics extraction using topological and LiDAR data

➔ **Automatic extraction of the characteristics**

Outcome : automated PV registry

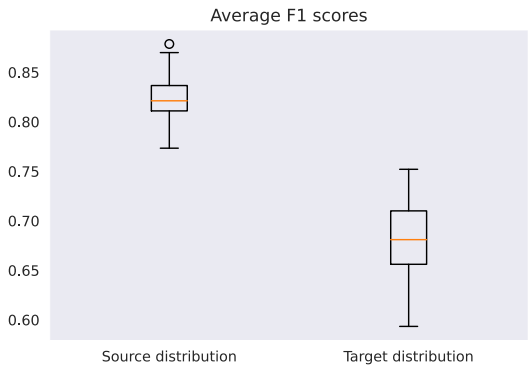
➔ **Aggregated and standardized data**

➔ Expected to be the **largest registry** with this level of detail

Scientific questions

- Performance of current models are worse on France than on their source data. Issue: **sensitivity to domain shift** (which is widespread in machine learning)

- We need to be able to **certify the method**:
  1. We need to <u>understand how domain shift affects a model's accuracy.</u>
  2. We need to <u>assess and investigate the accuracy</u> over the whole deployment area.

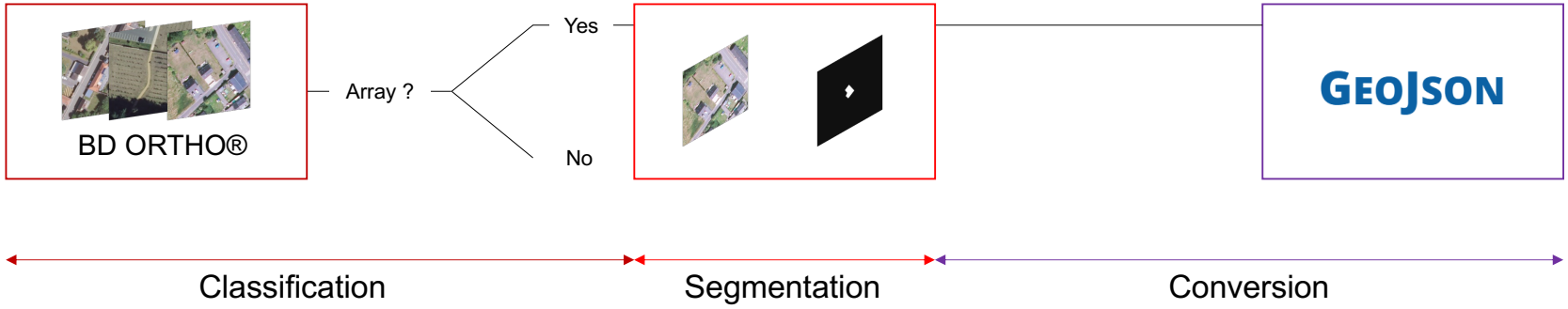


Performance drop from one region to the other (metric : F1 score)

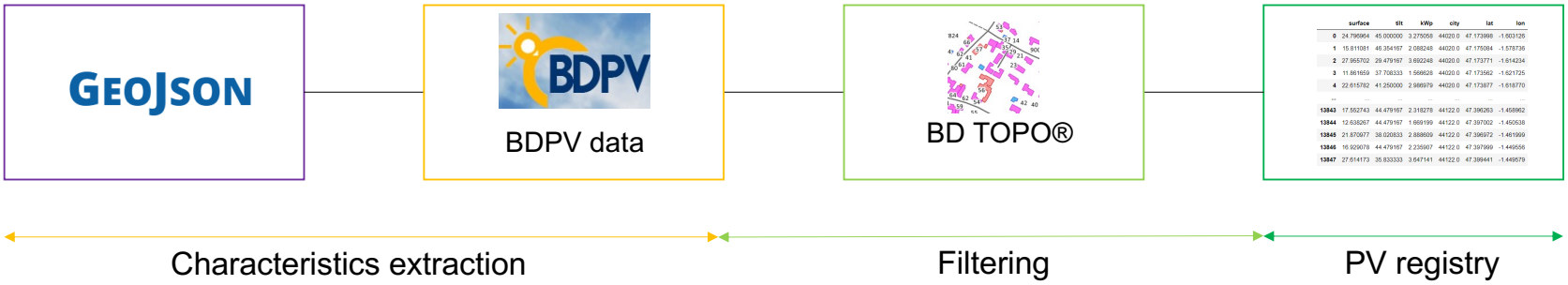
Some false positives

The automated PV registry pipeline

1. Arrays segmentation



2. Characteristics extraction and filtering



	surface	tilt	kWp	city	lat	lon
0	24.796964	45.000000	3.275058	44020.0	47.173986	-1.603126
1	15.811081	46.354167	2.088248	44020.0	47.175094	-1.578736
2	27.965702	29.479167	3.692248	44020.0	47.173771	-1.614234
3	11.861659	37.708333	1.566628	44020.0	47.173562	-1.621725
4	22.615782	41.250000	2.986979	44020.0	47.173877	-1.618770
...	...	...	...	...	...	...
13843	17.552743	44.479167	2.318278	44122.0	47.396263	-1.458962
13844	12.638267	44.479167	1.669199	44122.0	47.397002	-1.450538
13845	21.870977	38.020833	2.888609	44122.0	47.396972	-1.461999
13846	16.929078	44.479167	2.235907	44122.0	47.397999	-1.449556
13847	27.514173	35.833333	3.647141	44122.0	47.399441	-1.449579

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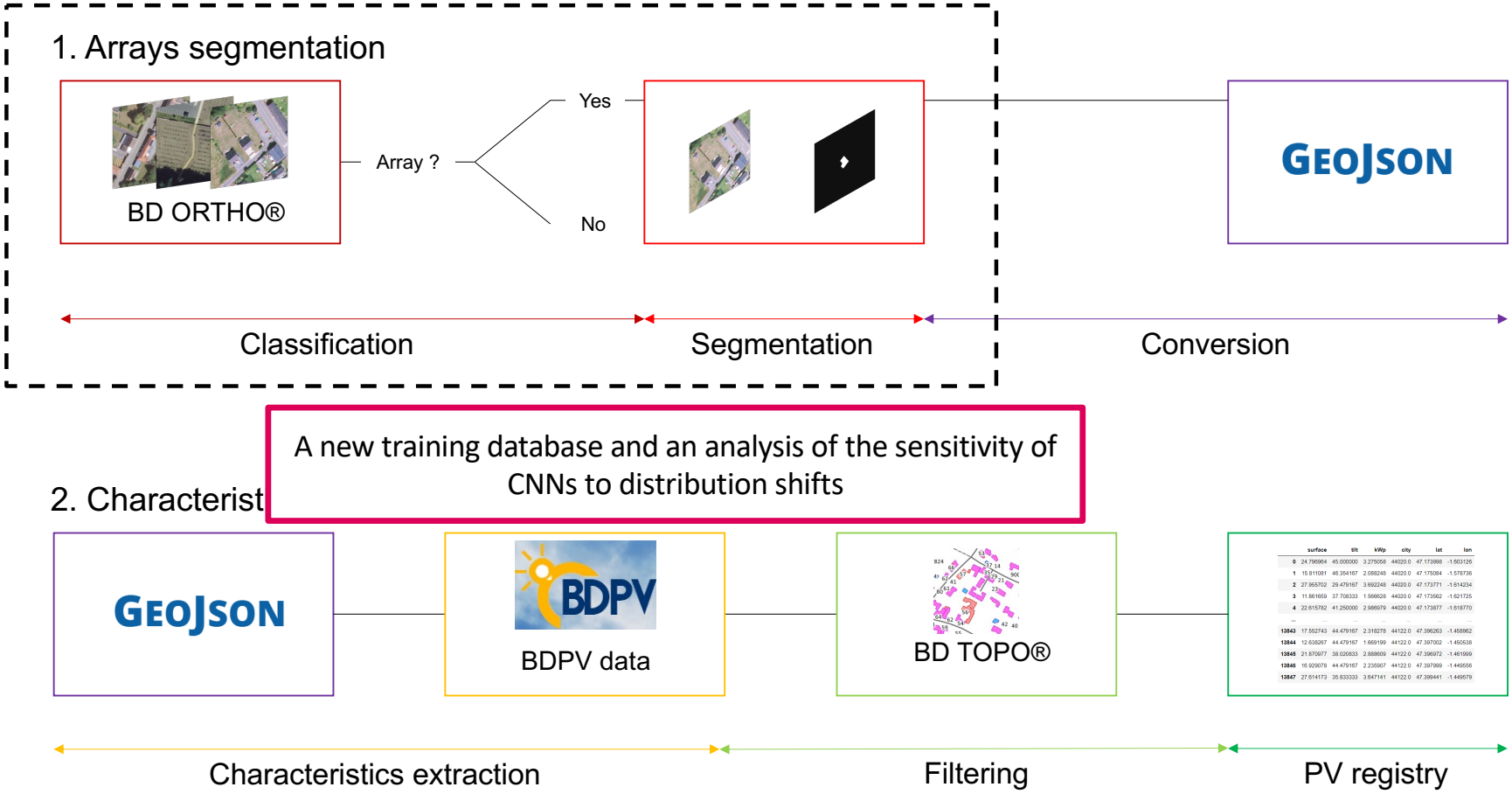

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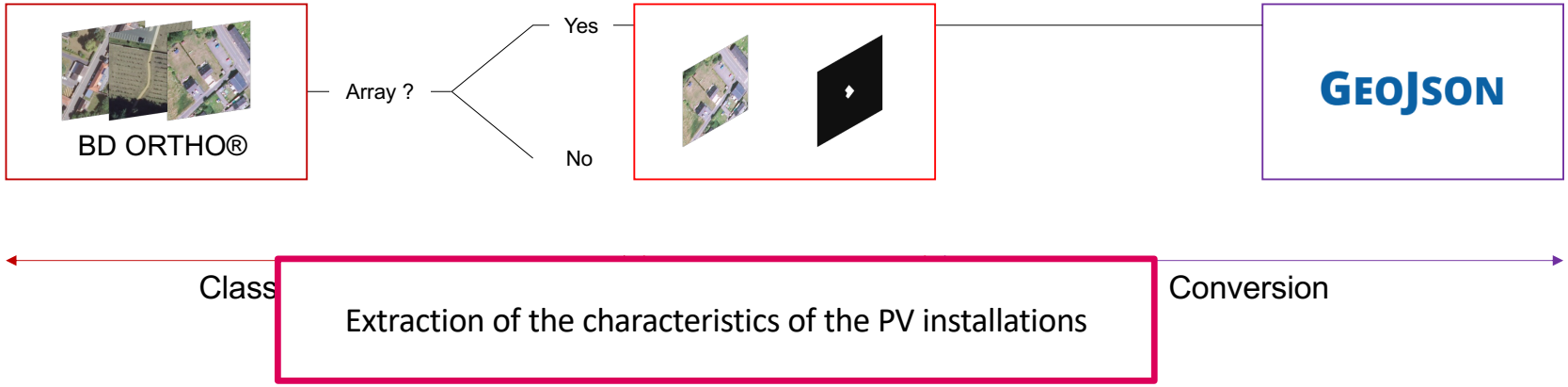

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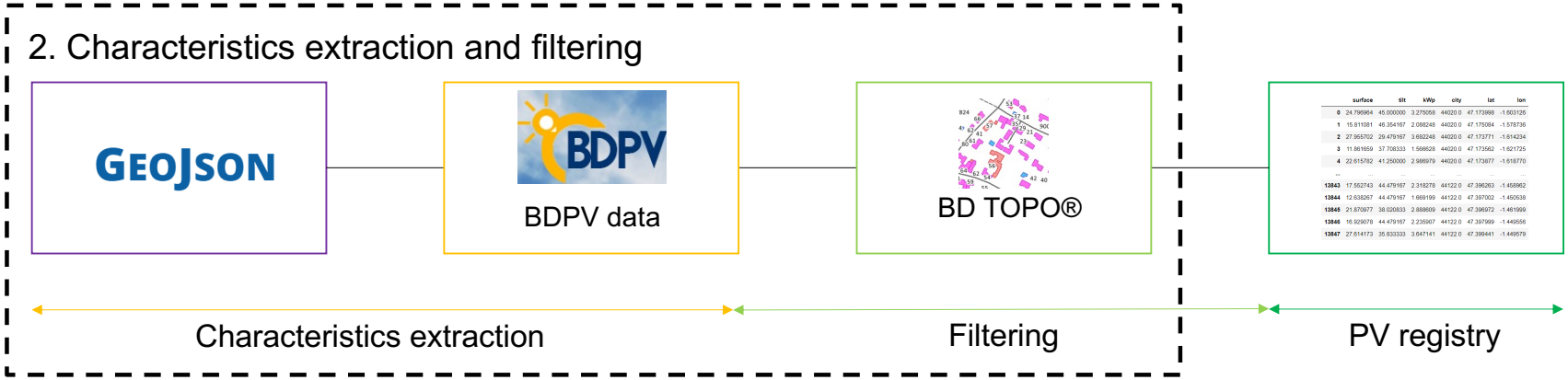

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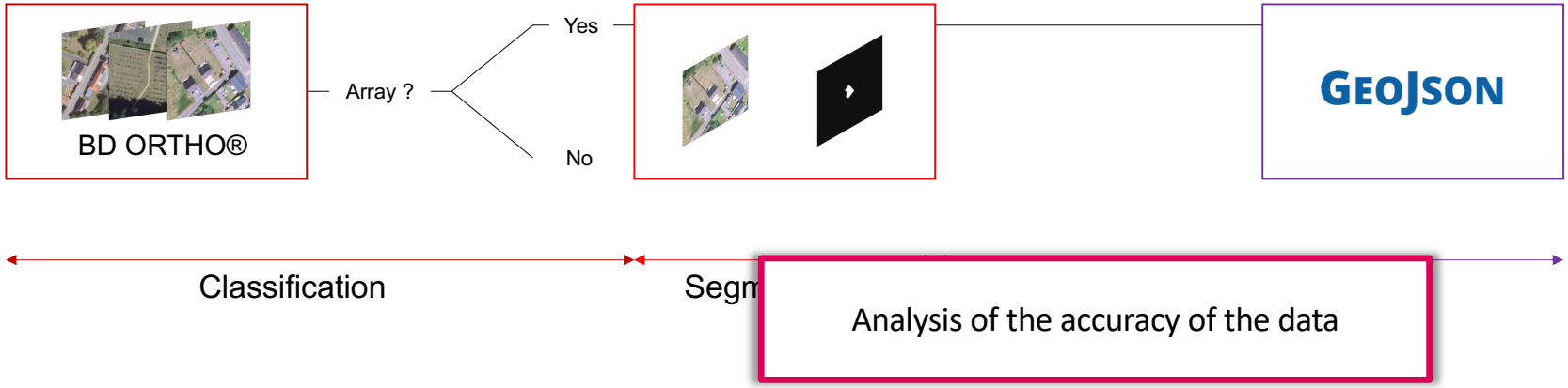

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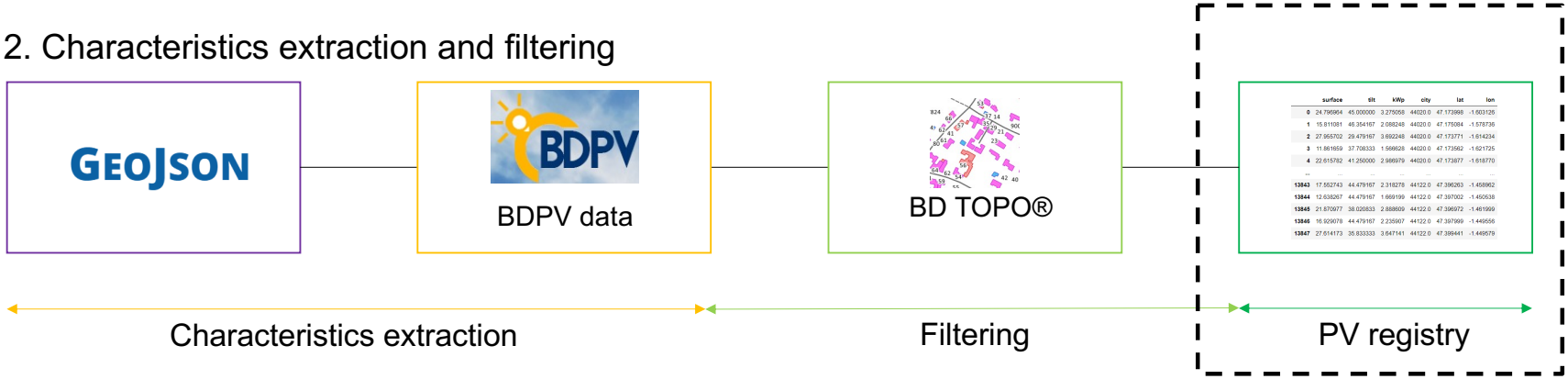

2 Accuracy assessment

Large scale assessement of the accuracy of the registry

	surface	tilt	kWp	city	lat	lon
0	24.796964	45.000000	3.275058	44020.0	47.173998	-1.603126
1	15.811081	46.354167	2.088248	44020.0	47.175084	-1.578736
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PV registry

- **Problem:** standard metrics (accuracy, F1, IoU,…) have **no meaning from the application standpoint**
- **Proposed approach : re-aggregate** the characteristics and compare them with the smallest scale available, the national registry of installations (*Registre national d'installations – RNI*)
- The RNI contains the **aggregated installed capacity** and the **number of installations** for installations that have an **installed capacity lower than 36 kWp** at the **scale of the cities**.

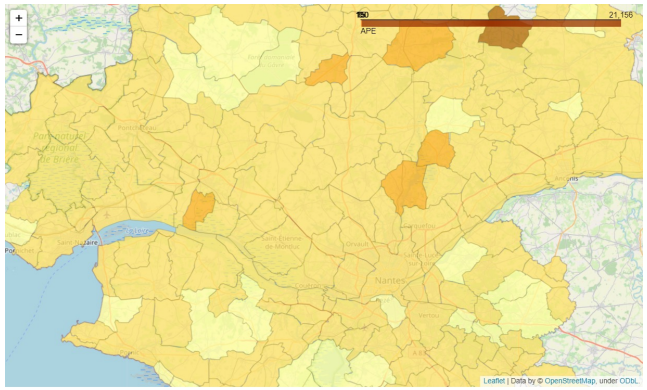
It allows us to **satisfy certifiability constraints:**

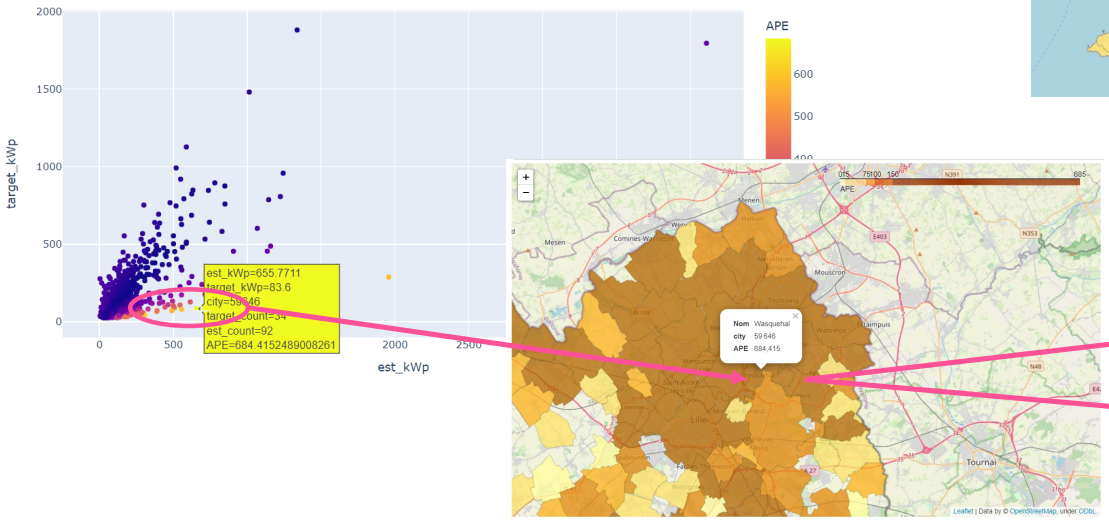
1. Control the accuracy over the whole territory
   RNI is available everywhere in France
2. Outputs investigation
   Tools can be developped to visualize the results

Large scale assessement of the accuracy of the registry

- « End user » accuracy example: average percentage error of the installed capacity, per city

➔ Identify « pathological cases »





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Thank you !

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